Kernels

- Kernels Represent a **dot product of two vectors** in a higher dimensional feature space
- Any Machine Learning Model that contains dot product of input samples can be kernelized
- Instead of calculating dot product between input vectors, calculate dot product between input vectors after mapping them to higher dimensional space

$$a \cdot b \rightarrow \text{map to } \varphi(a) \text{ and } \varphi(b) \rightarrow \varphi(a) \cdot \varphi(b) = K(a, b)$$

- Different Kernel functions are based on different mapping functions
- *Calculations of mapping and calculations of dot product in higher dimensional space are very expensive.*
- Use **Kernel Trick** to minimize both calculations (at no extra cost)

